

Team mates: Anantharaman, Bhuvana, Mohammed Ghouse

Domain: SDE in IT

Start Date: 02-03-2026

End Date: 25-03-2026

Demo Date: 28-03-2026, 3:30 PM

Project Problem Statement

Project Title: Personal Task Manager Web Application (Full-Stack SDE Project)

In the software industry, an SDE (Software Development Engineer) is expected to build applications that solve real-life problems, handle data properly, and provide a smooth user experience. One of the most common and useful applications in daily life is a **Task Manager**. People often struggle to organize their work, remember deadlines, and plan activities. A proper task manager can make life more productive and reduce confusion.

However, most students who want to become SDEs only practice coding problems but rarely build **complete applications** from start to finish. They do not get enough experience in working with frontend, backend, API design, database interactions, authentication, and user interface—all of which are needed in the IT industry. This project helps solve that gap.

The goal of this project is to build a simple but professional-style **Personal Task Manager Web App**. The student will learn how to create a full-stack application where users can create tasks, edit them, mark tasks as completed, delete tasks, and organize tasks based on priority or deadline. The entire system will be built using **free and beginner-friendly tools**.

The backend will be developed using **Node.js with Express** (or Python FastAPI if preferred). The backend will store tasks in a lightweight database such as **MongoDB Atlas (Free Tier)** or **SQLite**. The backend must have clean API routes such as:

- `/tasks` (list all tasks)
- `/tasks/add` (add new task)
- `/tasks/update/:id`
- `/tasks/delete/:id`

The frontend will be built using **React**, which is the most commonly used library in modern SDE roles. The user interface must be clean, easy to use, and should allow users to perform all actions without refreshing the page.

Students will also learn **state management**, **API integration**, and **basic error handling**. They will understand how to structure folders and files, how to write reusable components, and how a real SDE workflow feels. The project will give them the confidence to work with real company-style tasks in the future.

This project also teaches important habits such as writing documentation, maintaining code cleanliness, and preparing for a demo like in real IT companies. By the end, the student will understand the fundamentals of full-stack development, API handling, CRUD operations, UI/UX basics, and good engineering practices—all essential skills to become a successful Software Development Engineer.

WEEK-WISE PLAN

WEEK 1 — Backend API Development

Goal: Build the backend server that handles all task operations.

Tasks:

1. **Set Up Backend**
 - Install Node.js
 - Create Express server
 - Create folder structure (`routes`, `controllers`, `models`)
2. **Create REST API Endpoints**
 - Add Task
 - Get all Tasks
 - Update Task
 - Delete Task
3. **Database Setup**
 - Use MongoDB Atlas Free Tier or SQLite
 - Create Task model: `{ title, description, deadline, priority, completed }`

Week 1 Output:

✓ Backend fully working with all CRUD operations

WEEK 2 — Frontend UI (React)

Goal: Create a clean task management interface.

Tasks:

1. **React Setup**
 - Create pages and components
 - Navbar + Task List + Add Task Form
2. **Connect Frontend with Backend**
 - Use Axios to fetch tasks
 - Display tasks in a neat card or list
3. **UI Improvements**
 - Add “Mark as Done” button
 - Show completed tasks differently

Week 2 Output:

- ✓ Working UI
 - ✓ Data loaded from backend
-

WEEK 3 — Additional Features + Polishing

Goal: Make the app feel more like a real SDE project.

Tasks:

1. **Extra Features**
 - Search bar
 - Filter tasks (priority, deadline, completed)
2. **Error Handling**
 - Show message if something fails (ex: “Cannot connect to server”)
3. **Local Storage**
 - Save user preferences like theme (light/dark mode)

Week 3 Output:

- ✓ Polished app
 - ✓ Extra filters
 - ✓ Better user experience
-

WEEK 4 — Final Presentation & Demo

Tasks:

1. **Create PPT**
 - Problem
 - Approach
 - Architecture diagram
 - Screenshots
 - Learnings
2. **Live Demo**
 - Add, edit, delete, and filter tasks
 - Explain backend & frontend flow
3. **Q&A Preparation**
 - What is an API?
 - What is CRUD?
 - What is React state?
 - Why use Express?

Week 4 Output:

- ✓ Professional demo ready
- ✓ Clear explanation
- ✓ Full-stack project completed